

Scalability & Future Plan

Date	03 October 2026
Team ID	6ab7f3e4b8bf817b7f724e1d
Project Name	FitBuddy
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local SQLite Database	Limited concurrency handling and scalability for large enterprise-level user traffic.	Medium
2	Synchronous API Rate Limits	Heavy bursts of AI requests to Google Gemini could face rate limits or latency bottlenecks.	High
3	Lack of User Authentication	Absence of secure user accounts/passwords means user history is tied only to local sessions or simple ID lookups.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-server local FastAPI instance (uvicorn).	Scale horizontally using Docker containers, Kubernetes, and an Nginx load balancer to manage high concurrent traffic.
2	Data Storage	Local SQLite database file (fitbuddy.db).	Migrate to a robust relational database management system like PostgreSQL or MySQL hosted on a cloud instance.
3	Performance	Direct synchronous Gemini API calls.	Implement background task queues using Celery and Redis to handle plan generations asynchronously without blocking user requests.
4	Security	Minimal environment variable configuration (.env).	Introduce OAuth2 authentication, JSON Web Tokens (JWT), and role-based access control (RBAC) for admins and users.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Authentication & Profiles (JWT login, saved workout history, progress tracking charts)	Q3 2026	Enhances user retention and provides personalized long-term fitness tracking.
Phase 3	Wearable Device Integration (Sync data from Apple Health, Google Fit, or smartwatches)	Q4 2026	Allows AI to automatically adapt workout intensity based on real-time biometric telemetry.
Phase 4	Community & Social Sharing (Share custom plans, community leaderboards, trainer reviews)	Q1 2027	Transforms FitBuddy from an individual tool into an engaging interactive social fitness ecosystem.